

Biology — Checklist Marking Rubric

Examiner Marking Guidelines & Scoring Rubric — Checklist Edition (atomic, evidence-based criteria for consistent AI/human marking)

TOTAL PAPER MARKS: 40

1. General Principles & Examination Rules

- **Strict Evaluation:** Award marks strictly based on scientific accuracy, clarity, and logical consistency.
- **Scoring Breakdown:** Full marks on a criterion require a complete, scientifically accurate, and well-structured explanation. A criterion is awarded 0 for incorrect scientific knowledge, irrelevant answers, or complete omissions.
- **Step-by-Step Marking:** Examiners must mark each part step-by-step, evaluating the learner's response against the specific criteria below rather than granting holistic impressions.
- All 2 items are compulsory. Do not award marks for unanswered items.
- Be consistent throughout for all students, strictly applying the same rules throughout. If a particular answer was marked incorrect for one student, mark the same answer incorrect for every other student — never change standard mid-session.

The RACE Competency Framework (apply across every task/part): R – Relevance (addresses the specific prompt, scenario context, and biology concepts directly); A – Accuracy (correct identification of scientific terms, processes, definitions, causal relationships); C – Coherence (logical flow of thought, clear structure, correct use of language and grammar); E – Excellence (exceptional insight, detailed biological mechanics, meets all criteria thoroughly). The checklist criteria below already encode Relevance and Accuracy per line; Coherence and Excellence should be treated as a light modifier only within a criterion that is otherwise satisfied (i.e. do not invent extra marks for style beyond the fixed subtotal for each part).

How to use this checklist: each part below is broken into small, independently checkable criteria. Award the stated mark ONLY when the student's answer contains explicit, correct evidence of that specific criterion (a fact, mechanism, named example, or reasoning step). Do not award a criterion for a vague or generic statement that merely gestures at the topic. Partial understanding of a criterion still receives 0 for that criterion — there is no partial credit within a single checklist line; partial credit across a part comes from awarding some lines and not others. Apply every line identically to every student.

Item 1: Nutrition, Kwashiorkor, and Digestion (Total: 20 Marks)

Identification [2 Marks]

Award mark(s) if the answer...	Marks
correctly names the condition as Kwashiorkor	1
correctly identifies it as protein-energy malnutrition / a protein-deficiency disease (not merely naming it)	1
Subtotal	2

Full-Marks Exemplar (model answer — preserved from source rubric)

Condition Name: Kwashiorkor. Kwashiorkor is a severe form of protein-energy malnutrition that occurs when a child's diet consists almost entirely of carbohydrates (such as cassava porridge and sweet potatoes) without sufficient protein intake.

Zero / Low-Marks Exemplar (incorrect answer — preserved from source rubric)

The children are suffering from Marasmus or bewitchment. The children are weak because cassava and sweet potatoes are bad foods that poison the stomach.

Examiner Note: Incorrect diagnosis (Marasmus/witchcraft) and non-scientific rationale. Award 0 on every criterion line.

Explanation of Damage [6 Marks]

Award mark(s) if the answer...	Marks
explains edema/swollen belly correctly by naming reduced blood plasma protein (e.g. albumin) and lowered osmotic pressure as the cause of fluid leaking into tissues	2
explains muscle wasting / impaired growth via lack of amino acids for tissue building and repair	1
explains hair/skin pigmentation change via lack of amino acids (e.g. tyrosine) needed for melanin synthesis	1
explains anemia via impaired hemoglobin/red blood cell production from protein deficiency	1
gives a further correct, distinct physiological consequence of protein deficiency not already credited above (e.g. weakened immune response, stunted growth generally)	1
Subtotal	6

Full-Marks Exemplar (model answer — preserved from source rubric)

When a child eats only starchy foods:

1. Fluid Imbalance (Edema): Protein is essential for synthesizing blood plasma proteins like albumin. A lack of protein reduces blood osmotic pressure, causing plasma fluid to leak into surrounding interstitial tissues. This leads to water retention, causing the characteristic swollen belly (ascites) and edema in limbs.
2. Impaired Growth & Muscle Wasting: Proteins provide amino acids necessary for building cellular structures, tissue repair, and muscle growth. Without amino acids, body proteins are broken down to sustain vital organs, leading to weak bodies and stunted growth.

3. Pigmentation & Skin Damage: Melanin (the pigment responsible for skin and hair color) requires amino acids like tyrosine for synthesis. A deficiency leads to reddish/lightened hair and pale, flaky skin.
4. Anemia: Proteins are vital for hemoglobin formation. Insufficient protein impairs red blood cell production, resulting in pale skin due to anemia.

Zero / Low-Marks Exemplar (incorrect answer — preserved from source rubric)

When a child stops breastfeeding, their stomach cannot handle cassava, so it boils inside and turns the hair red. They should drink traditional herbs to cure the belly.

Examiner Note: Non-scientific explanation with no mention of albumin, osmotic pressure, amino acids, or melanin. Award 0 on every criterion line.

Importance of Protein in Children [6 Marks]

Award mark(s) if the answer...	Marks
growth & structural development: amino acids build cellular structures/muscles/bones during rapid cell division	1
enzyme synthesis: proteins act as biological catalysts for metabolic processes	1
hormone synthesis: proteins/amino acids form chemical messengers (e.g. growth hormone, insulin)	1
immune function: antibodies that fight infection are proteins	1
tissue repair: proteins repair daily wear-and-tear damage to body tissues	1
correctly explains why children specifically need proportionally more protein than adults (rapid growth/cell division demand)	1
Subtotal	6

Full-Marks Exemplar (model answer — preserved from source rubric)

Why Growing Children Need Protein Foods (6 Marks):

1. Growth and Structural Development: children undergo rapid cell division; proteins supply amino acids required to synthesize cellular membranes, structural tissues, muscles, and bones.
2. Enzyme and Hormone Synthesis: biological catalysts (enzymes) and chemical messengers (such as growth hormone and insulin) are proteins needed for metabolic processes.
3. Immune System Function: antibodies that fight infections are proteins. Protein-deficient children are susceptible to frequent illnesses.
4. Tissue Repair: daily activity leads to tissue wear and tear; proteins are necessary to repair damaged body tissues.

Zero / Low-Marks Exemplar (incorrect answer — preserved from source rubric)

Children need protein so they don't die.

Examiner Note: No biological mechanism given — no mention of growth, enzymes, hormones, immunity, or repair. Award 0 on every criterion line.

Digestion of Proteins [6 Marks]

Award mark(s) if the answer...	Marks
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stomach: correctly describes HCl activating pepsinogen into pepsin in an acidic environment (~pH 1.5–2.0)	1
stomach: correctly states pepsin breaks long protein chains into smaller polypeptides	1
small intestine: correctly states pancreatic trypsin is released into the duodenum in an alkaline environment (bile/bicarbonate)	1
small intestine: correctly states trypsin digests polypeptides into shorter peptides	1
small intestine: correctly states peptidases/erepsin in the ileum complete digestion into single amino acids	1
absorption: correctly states amino acids are absorbed across the villi of the ileum into the bloodstream	1
Subtotal	6

Full-Marks Exemplar (model answer — preserved from source rubric)

Contrary to the belief that young children cannot digest thick protein foods like groundnut paste, a child's digestive system produces specific enzymes tailored to break proteins down into simple, absorbable amino acids. Stomach: gastric juice contains hydrochloric acid (HCl) and inactive pepsinogen; HCl creates an acidic pH (~1.5–2.0), activating pepsinogen into pepsin, which breaks long protein chains down into smaller polypeptides. Small Intestine (Duodenum & Ileum): pancreatic juice releases trypsin into the duodenum (in an alkaline pH maintained by bile/bicarbonate), which digests polypeptides into shorter peptides; epithelial glands in the ileum then secrete peptidases (erepsin), completing digestion by breaking peptides down into single amino acids. Absorption: the resulting amino acids are absorbed across the villi of the ileum into the bloodstream via active transport/diffusion and delivered to cells throughout the body.

Zero / Low-Marks Exemplar (incorrect answer — preserved from source rubric)

The stomach cannot break down groundnut paste because groundnut paste is too thick and clogs the intestine. The stomach juice melts cassava but groundnuts just sit in the stomach and cause pain.

Examiner Note: Fails to describe enzyme actions (pepsin, trypsin, peptidases), provides incorrect digestive mechanics, and validates false cultural myths without scientific basis. Award 0 on every criterion line.

Item 2: Water Movement in Plants & Osmosis (Total: 20 Marks)

Osmotic Mechanism [8 Marks]

Award mark(s) if the answer...	Marks
correctly identifies that dry fertilizer plus salty water creates a hypertonic soil solution (high solute concentration)	1
correctly uses water/solute potential terminology (Ψ s / Ψ_w) rather than vague language	1
correctly explains the water potential gradient: soil water potential lower than root hair cell sap	1
correctly explains that water moves from a region of higher water potential to a region of lower water potential	1
correctly names the process as exosmosis/osmosis specifically (not just “water movement”)	1
correctly states the movement occurs across the semi-permeable root cell membrane	1
correctly explains loss of turgor pressure and/or plasmolysis as the cell vacuole shrinks	1
correctly links the mechanism to the observed wilting despite physically wet soil (the key counter-intuitive insight)	1
Subtotal	8

Full-Marks Exemplar (model answer — preserved from source rubric)

1. Creation of a Hypertonic Solution: applying a large quantity of dry artificial fertilizer combined with salty water significantly increases the concentration of solute ions (salts and nitrates) in the soil solution.
2. Water Potential Gradient: the soil water becomes hypertonic relative to the cell sap inside the root hair cells — the soil solution has a lower water potential (Ψ) / higher solute concentration than the internal environment of the plant's root hair cells.
3. Exosmosis: water moves along a water potential gradient from a region of higher water potential (inside the root hair cells) to a region of lower water potential (the hypertonic soil) across the semi-permeable cell membrane via osmosis (exosmosis).
4. Loss of Turgidity: as water rapidly leaves the root cells, the vacuole shrinks, and the plasma membrane pulls away from the cell wall (plasmolysis).
5. Wilting: root cells lose turgor pressure, which normally provides mechanical support to non-woody stem tissues. As a result, the stems and leaves lose firmness and droop/wilt, despite the soil being physically wet.

Zero / Low-Marks Exemplar (incorrect answer — preserved from source rubric)

The fertilizer burned the roots because it was too hot. Salty water blocks the holes in the roots so water cannot go inside the plant. The water stayed in the soil because the fertilizer sucked it up like a sponge.

Examiner Note: Uses non-scientific concepts (“burned,” “blocked holes,” “sucked up like a sponge”) without referencing osmosis, water potential, or hypertonic solutions. Award 0 on every criterion line.

Effect of Leaf Spraying [6 Marks]

Award mark(s) if the answer...	Marks
identifies that the original problem was osmotic desiccation, not pest attack (misdiagnosis)	1
explains stomatal blockage or damage from the chemical spray	1
explains direct chemical toxicity/damage to leaf epidermal or mesophyll tissue	1
explains additional hypertonic stress from solutes deposited on the leaf surface drawing out more moisture	1
explains disruption of transpiration/gas exchange from stomata being forced shut or guard cells destroyed	1
explains that this accelerates desiccation/necrosis (tissue death) rather than curing the plant	1
Subtotal	6

Full-Marks Exemplar (model answer — preserved from source rubric)

Why Spraying Leaves Made Damage Worse (6 Marks):

1. Misdiagnosis & Additional Stress: the plant was suffering from osmotic desiccation (water loss), not pest attack.
2. Stomatal Blockage & Direct Leaf Toxicity: spraying high concentrations of strong chemicals directly on wilted leaves damages the fragile epidermal and mesophyll tissues.
3. Chemical Hypertonicity: the chemical spray deposited additional solutes on leaf surfaces, drawing remaining moisture out of leaf epidermal cells via osmosis.
4. Disruption of Transpiration & Gas Exchange: toxic chemical build-up can force stomata shut or destroy guard cells, preventing photosynthesis and gas exchange while accelerating tissue necrosis (drying up and death).

Zero / Low-Marks Exemplar (incorrect answer — preserved from source rubric)

The spray killed the leaves because the farmer sprayed during the day instead of at night.

Examiner Note: Fails to explain leaf desiccation mechanism and misunderstands the actual cause. Award 0 on every criterion line.

Correct Farming Advice [6 Marks]

Award mark(s) if the answer...	Marks
advises using fresh, non-saline water rather than saline deep-well water	1
advises applying fertilizer at a controlled, recommended quantity (avoid over-application)	1
advises correct placement of fertilizer away from the stem (e.g. in a ring around the drip line)	1
advises irrigating/dissolving fertilizer properly to dilute solute concentration	1
advises accurate diagnosis (checking soil moisture/root condition) before applying any chemical treatment	1
advises a further genuinely practical safeguard (e.g. soil testing before fertilizing, staged/split fertilizer application)	1

Subtotal	6
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Full-Marks Exemplar (model answer — preserved from source rubric)

Correct Watering and Fertilization Advice for the Future (6 Marks):

1. Use Fresh, Non-Salty Water: water crops using fresh river or rainwater rather than saline deep-well water to prevent salt accumulation in the soil.
2. Proper Fertilizer Application Rate: apply artificial fertilizers in strictly controlled, recommended quantities; avoid excessive application.
3. Placement: apply dry fertilizer a few centimeters away from the plant stem (e.g. in a ring around the drip line) rather than directly against the stem, preventing localized high salt concentrations.
4. Dissolve/Irrigate Appropriately: apply fertilizer when soil is already moist or thoroughly irrigate with clean water immediately after application to dilute solute concentration and facilitate balanced absorption.
5. Accurate Problem Diagnosis: check soil moisture and root osmotic conditions before spraying pesticides unnecessarily.

Zero / Low-Marks Exemplar (incorrect answer — preserved from source rubric)

Next time, the farmer should use more salty water to wash away the chemical spray. He should put fertilizer directly on top of the leaves next time.

Examiner Note: Contains scientifically detrimental advice and misunderstands proper agricultural methods. Award 0 on every criterion line.